Relief Forge

The basics. One picture in, one carvable relief out.

CNC · Fibre laser · 3D print · Moulds · Coins & medals

Quick start guide · October 2026 · updated

snazzygaz.github.io/DepthNormal

Runs in your web browser · nothing is uploaded anywhere

THE GOAL

What this thing does

Relief Forge turns a **picture** into a **height map**: a grey image where **white = high** and **black = low**. CNC routers, fibre lasers and 3D printers read that grey as depth, so the picture comes out as a carved, raised object, like a coin, a medal or a stone panel.

Two AI models look at your picture. One guesses **how far away** everything is (depth). The other guesses **which way each bit of surface faces** (normals). The Forge blends the two into a clean relief and gives you sliders to shape it.

Everything happens on your own computer. Your pictures never leave it. The AI models download once (the first time) and are then kept by your browser. Your work is also saved in the browser about once a minute, so a crash or a closed tab doesn't lose it.

What you need

- **A computer with Google Chrome** (or Edge). A decent graphics card makes it fast. Without one it still works, just slowly.
- **About 1 GB of free space** for the AI models, and an internet connection for the first run.
- **A good picture.** This matters more than any setting:
  - High resolution, sharp, in focus.
  - The subject **cut out** from its background: a transparent background or one plain colour (all green, all white…).
  - Even, soft lighting. No hard shadows.
  - Crop to the subject. A small statue in a big scene wastes most of the depth on the scene.
  - AI image editing helps a lot: ask it to turn your picture into **plain white stone on a plain background**. This matters most for things with colour or shine (cars, clocks, animals). Flat art such as signs and lettering works fine as it is.

HOW IT FLOWS

The four steps

1

Setup

Load the two AI models. Only slow the first time.

2

Generate

Add your picture, press **Run passes**.

3

Sculpt

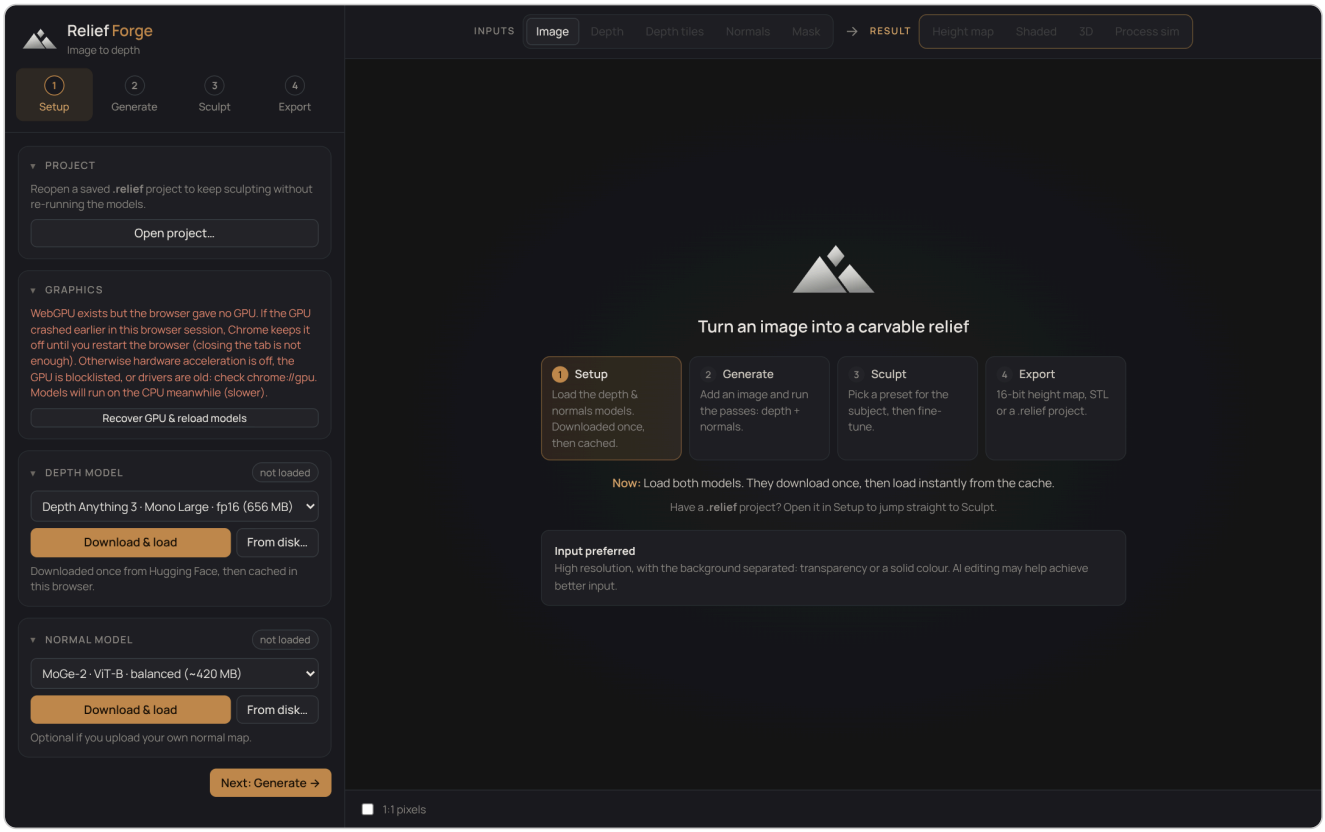
Pick a preset (or let it decide), nudge sliders, watch it update.

4

Export

Save the height map, an STL or the project.

The buttons for these four steps are at the top left. Do them in order. The big area on the right tells you what to do next (the **“Now:”** line).



The start screen. Left: the controls for the step you’re on. Right: what to do next. (This computer had no graphics card, so Graphics shows a warning.)

STEP 1

Setup

Restore last session NEW

Shows up when the browser has your last piece of work saved. One click and you're back where you were, settings and all.

Project

Already made something and saved it? **Open project…** loads it back and jumps straight to Sculpt. No AI needed.

Graphics

Tells you if your graphics card can be used. If it can't, everything still works on the processor, just slower. If the card crashes mid-job, a **Recover GPU** button appears here and next to Run: it restarts the models without restarting the browser.

Depth model

The AI that guesses distance. Press **Download & load**. Leave it on the first choice (Depth Anything 3) unless you know why you want another. Depth Anything 2 is also offered; it is **non-commercial only**.

Normal model

The AI that guesses surface direction. It gives all the crisp detail. Press **Download & load**. ViT-B is the sensible default; ViT-L is bigger and slower.

STEP 2

Generate

Source image

Drop your picture in, or click to choose it.

Working size

How big (in pixels) the relief is made. 2048 is a good default. Bigger = more detail, but slower.

Depth tiles / Normal tiles

The picture is cut into pieces, each piece is looked at closely, then they are stitched back together. **3 across** gives much more detail. Turn off for speed.

Normal detail (tokens)

How finely the normal AI looks at each piece. 2500 is the default. Higher is sharper but heavier on your graphics card, and can leave a faint blotchy pattern on flat areas.

Remove patch grid NEW

The normal AI leaves a faint waffle pattern on flat backgrounds. This finds it and takes it out. On by default; the note next to it says what it did.

Run passes

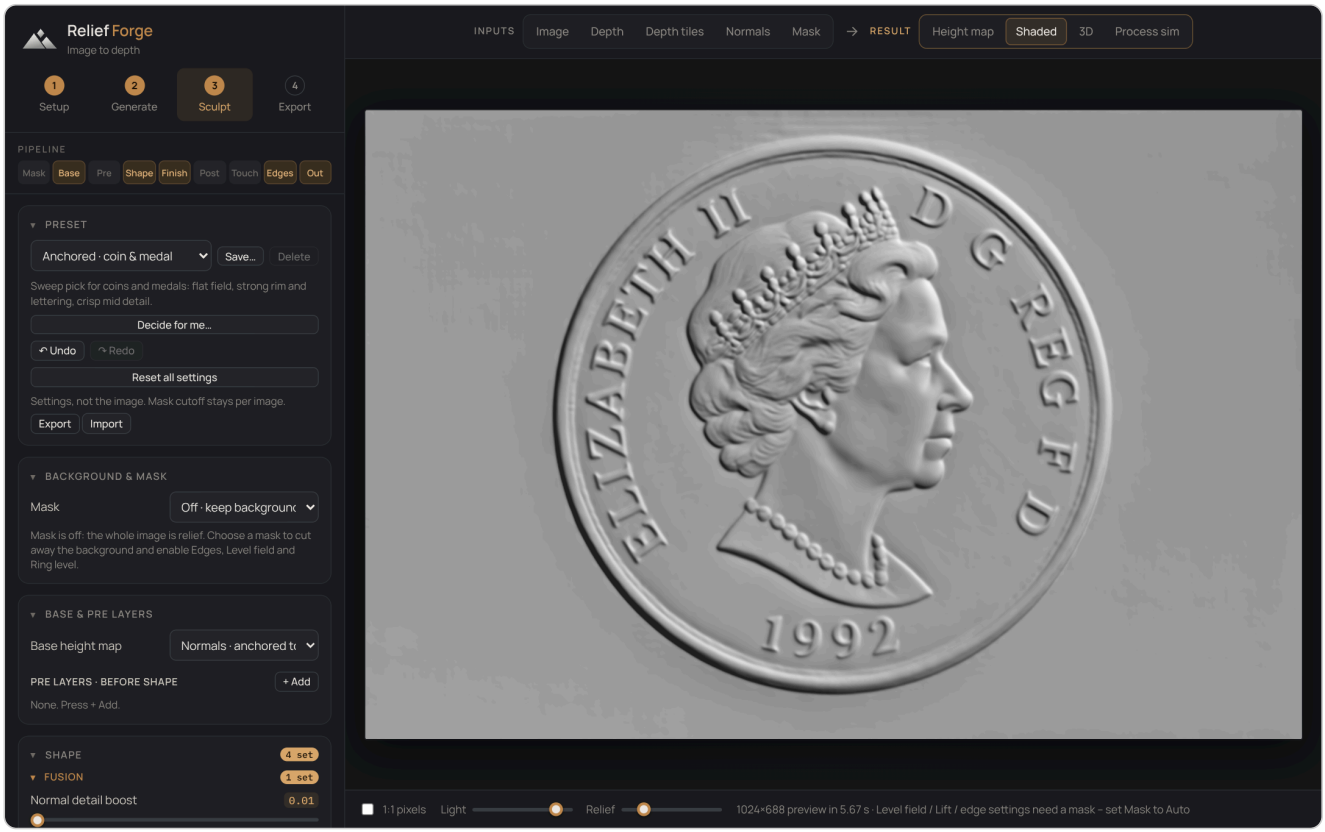
Press it and wait. When it says **Done**, go to Sculpt. A new picture starts on the **Anchored · all-rounder** preset.

LOOKING AT THINGS

The view tabs (top of the screen)

Inputs (left group) are the raw materials. **Result** (right group, in bronze) is what you'll actually get.

TAB	WHAT IT SHOWS
Image	Your picture.
Depth / Depth tiles	What the depth AI saw. White = near.
Normals	What the normal AI saw, as colours.
Mask	Red = background that gets thrown away. Clicking it also opens the mask settings and fades the rest, so you can focus; leave the tab and the panel goes back as it was.
Height map	**The actual output.** White = high.
Shaded	The height map lit from the side, so it looks like the real object. Easiest to judge.
3D	A real 3D model you can spin, in different materials.
Process sim NEW	What your CNC bit, 3D printer or laser can actually make of it, as a picture or in 3D.



The Sculpt step with the "Anchored · coin & medal" preset, Shaded view.

STEP 3

Sculpt, section by section

The panels run top to bottom in the order the work is done. Click a panel's title to fold it away. The **Pipeline** strip at the top lights up the stages you've changed; click one to jump to it.

Which preset?

Start in the **Recommended** group at the top of the list. These were tested on 22 projects (coins, statues, animals, machines, photos and stone renders). The older presets are still there further down, marked **Legacy**.

YOUR SUBJECT	PRESET
Not sure / anything	**Anchored · all-rounder**. Safe on everything. Or press **Decide for me…**
Coin, medal, badge, emblem	**Anchored · coin & medal**: flat field, strong rim and lettering.
Statue, figure, creature	**Anchored · keeps volume** for the most depth (can look soft on faces), or **Anchored · crisp contrast** for punchier modelling.
Scenes and textures	**Anchored · all-rounder** or **Crisp · classic base**.
Start from scratch	**Neutral · base only**, then build up yourself.

Preset

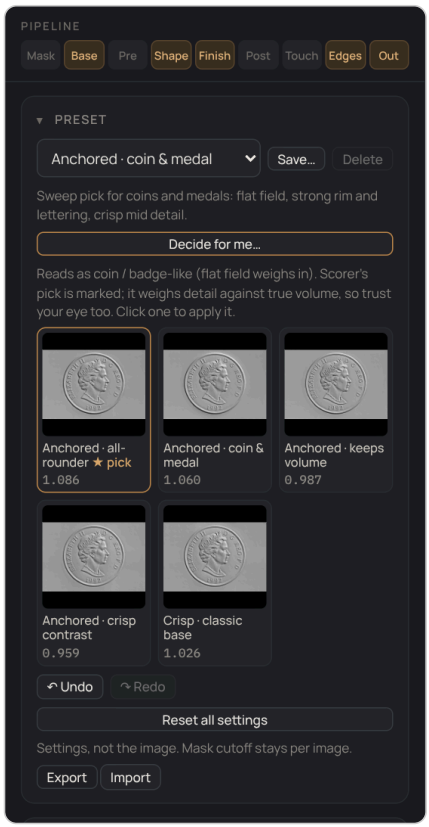
Start here. Pick one from the table. A note under the box says what it's for. **Reset all settings** puts everything back to plain. **Save…** keeps your own settings; **Export / Import** shares them as a file. Presets leave your edge settings alone.

Decide for me… NEW

Tries all five Recommended presets on your picture, scores them and shows them side by side with its pick starred. Click the one you like. Nothing changes until you do. Takes under a minute.

↶ Undo / ↷ Redo NEW

Steps back and forward through your setting changes. Ctrl+Z and Ctrl+Y work too.



Decide for me on a coin. It read the piece as coin-like and starred the all-rounder.

Background & mask

Decides what is subject and what is background. **Auto** usually gets it right. If not, press **Pick colour** and click the background. Paint with **Keep / Remove** to fix small mistakes. The background becomes flat black.

Base & pre layers

What the relief is built from. **Normals · anchored to depth** is the one the Recommended presets use: crisp detail from the normals, with the big shapes kept in the right order by the depth. **Depth anchoring size** sets where one hands over to the other. Pre layers let you mix in other maps; leave them alone at first.

Shape

The big moves: how much fine detail to add, how deep or flat the whole thing is, and levelling so it isn't tilted. **Level field** and **Auto-level tilt** make backgrounds and coin fields flat.

Finish

Clean-up and polish: **Denoise**, the **Detail EQ** (fine, medium, broad), **Crease deepen** for crisp lines, gamma and a tone curve.

Post layers

Blends another map over the finished result, e.g. a little texture. Optional.

Touch-up brush

Paint directly on the result: **Raise**, **Lower**, **Smooth**, **Erase**. Has its own Undo, and an on/off switch to compare.

Edges

How the subject meets the background. **Edge roll-off** makes it slope down softly instead of a cliff. **Lift subject** raises it off the floor like a slab (coins, badges).

Rim lock (coins) NEW

Makes a coin's rim exactly level all the way round: every point the same distance in from the edge gets the same height. **Double-click the slider** to find the rim by itself. Keep patterned bands (rope, lettering) outside it, as anything inside goes plain. Needs a mask.

Mould draft NEW

Slopes steep walls slightly (0–20°) so a cast comes out of the mould. Only adds material at the foot of walls; detail is never cut away.

Output

Stretch output uses the full black-to-white range. **Invert** turns it inside out (for a mould). **Mirror for mould** flips it so a cast reads the right way round.

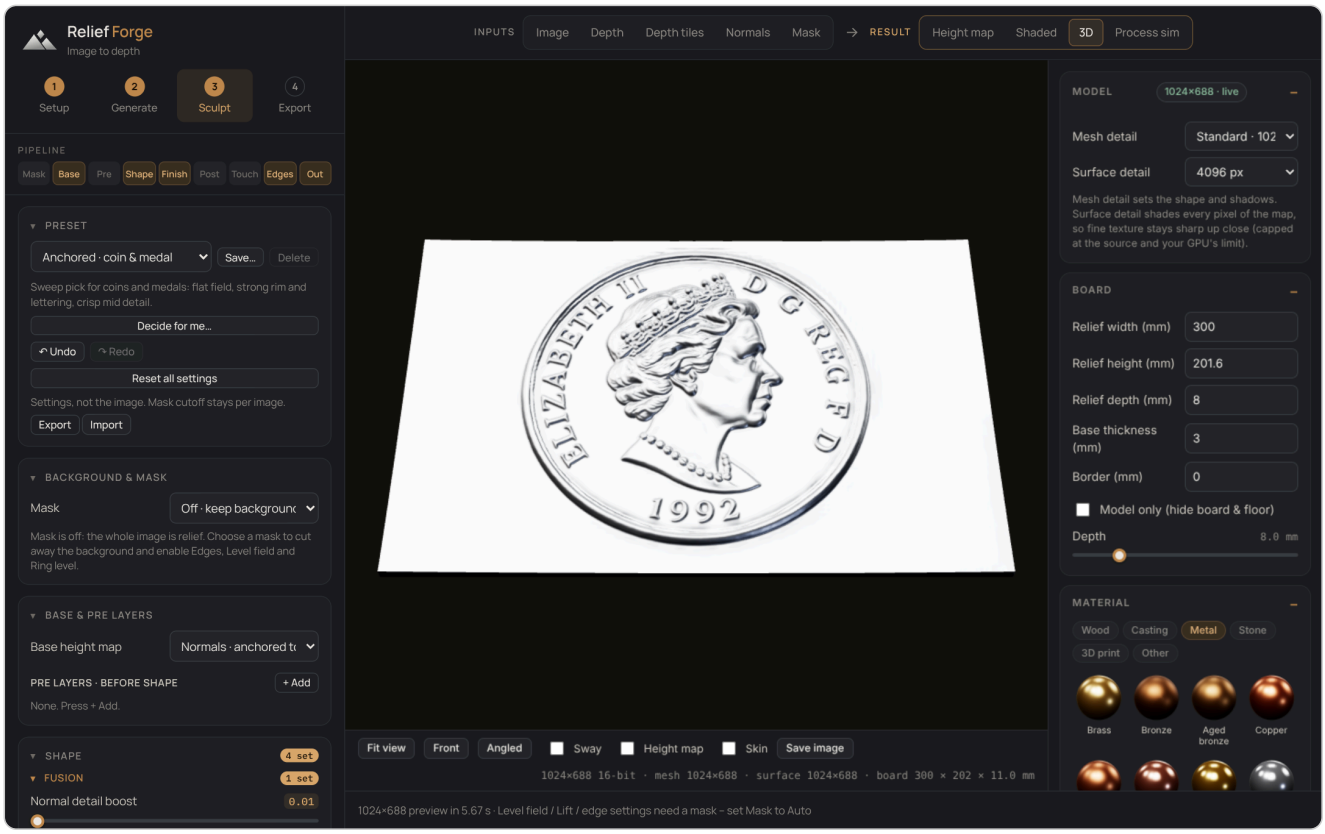
Advanced

Fine controls most people never need. Sizes in pixels also show millimetres, worked out from the width you set in Export.

Live updates while you drag. **Fast preview** works at half size while you tweak; exports are always full size. Hover any slider for a plain explanation of what it does.

CHECKING IT

The 3D view



The 3D tab, here in Steel. The model updates live as you change sliders.

Spin it

Drag to turn, scroll to zoom. **Fit view**, **Front** and **Angled** reset the camera.

Material

Click a ball to see it in steel, bronze, wood, stone, resin and more.

Board

Size and depth of the panel in millimetres, plus base thickness and border.

Light

Pick a lighting preset or move the light around.

Height map / Skin (bottom bar)

Height map shows the raw grey on the model. **Skin** wraps your original picture over it, to check things line up. Untick both to go back to the material.

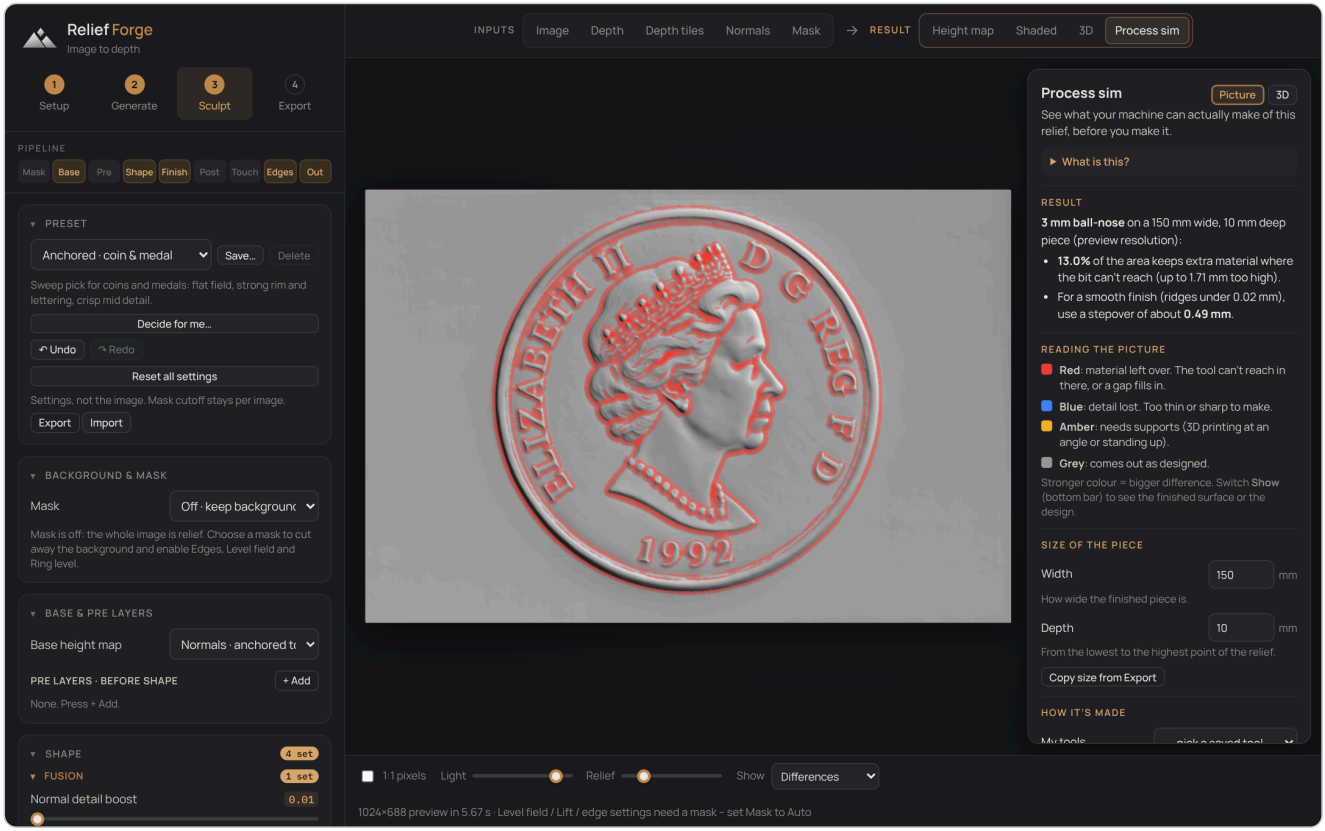
Save image

Saves a picture of the current view.

CHECKING IT NEW

Process sim: will my machine manage it?

A relief can hold detail no tool can make: a 3 mm bit can't reach into a 1 mm groove, and a printer can't print finer than its layers. The Process sim shows what your tool will really leave behind, before you waste wood or filament. It never changes your exports.

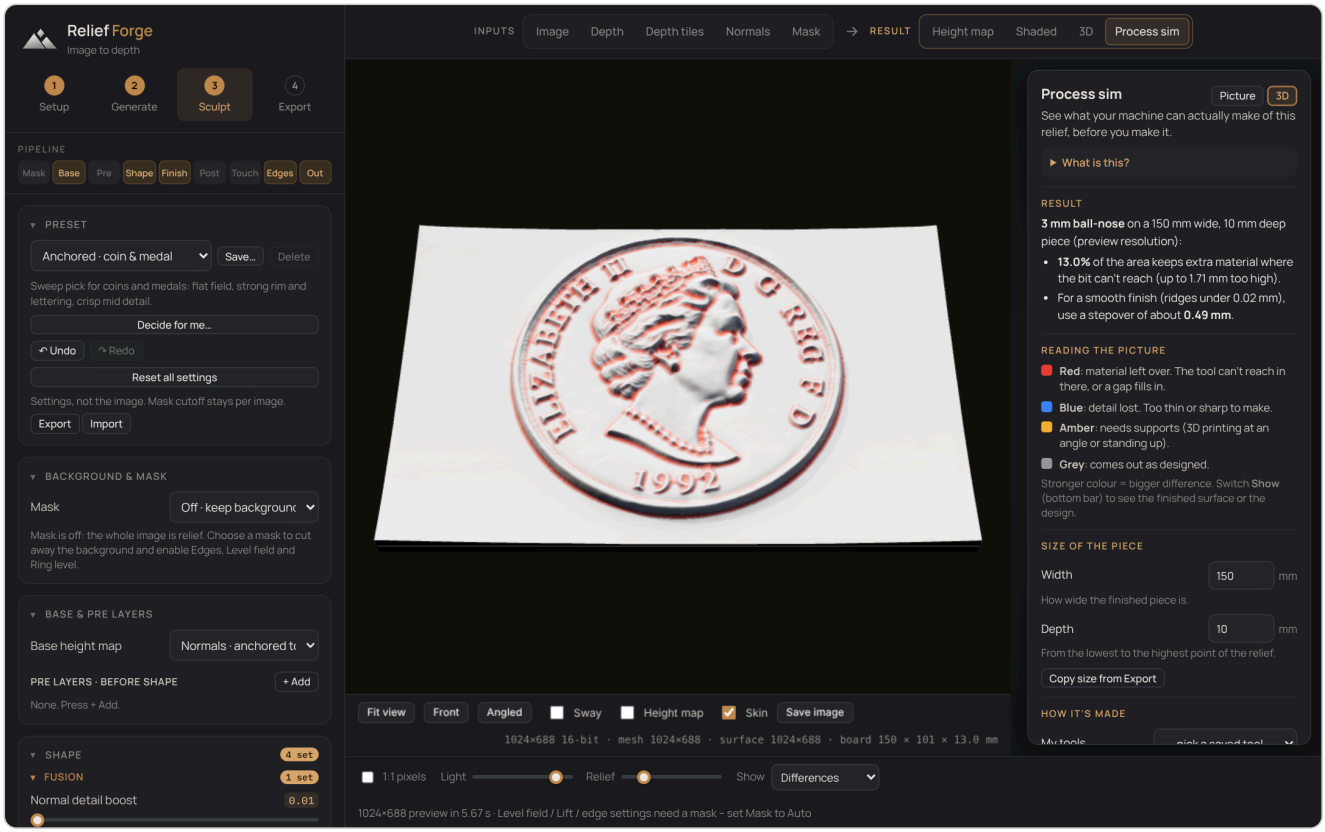


Process sim, Picture mode: a 3 mm ball-nose on a 150 mm coin. Red is where the bit can't reach.

Reading the picture

- **Red**: material left over. The tool can't reach in there, or a gap fills in.
- **Blue**: detail lost. Too thin or sharp to make.
- **Amber**: needs supports (3D printing at an angle or standing up).
- **Grey**: comes out as designed. Stronger colour = bigger difference.

The **Result** list in the panel says it in words: how much fills in, how much is lost, how tall the ridges between passes are, how many layers or passes it takes.



The same, switched to **3D**: the made surface at its real size, with the colours wrapped over it.

Picture / 3D

The switch at the top of the panel. **3D** shows the surface your tool would make, sized to the piece, and you can spin it like the 3D tab. **Show** (bottom bar) picks **Differences** (the colours), **As made** (bare surface) or **Design** (what you asked for).

Size of the piece

Width and depth in millimetres. Copied from Export the first time; **Copy size from Export** syncs it again.

How it's made

CNC: ball-nose, tapered ball-nose, V-bit, flat end mill or bull-nose, with its sizes and the **stepover** (gap between passes). It tells you the stepover for a smooth finish.
3D printing: filament or resin, layer height, line or pixel size, and **Print angle** (flat, tilted or standing up). Standing up gives finer steps on gentle slopes but may need supports.
Laser: fibre, diode or CO_2 fill in typical values; adjust spot size, wall angle and depth per pass.

My tools

Save the bit or printer you've set under a name and pick it again next time. **Export / Import** moves your list between computers as a file.

Red isn't always bad. Your CAM software can follow a big bit with a finer one for the details (a rest pass). Use the sim to see which areas need that finer bit.

STEP 4

Export

File name

The name used for everything you save. Leave blank for an automatic one.

Height map

Download 16-bit PNG: the main thing. Use this for CNC and lasers (smooth steps). 8-bit is only for programs that can't read 16-bit.

3D model · STL

A solid 3D file for printers and CAM. Set panel width, relief depth and backing. It can also build a **silicone pour box** around it and **split it** to fit your print bed.

Normal map

The colour surface map, for 3D artists.

Project

Save project keeps everything (picture, AI results, settings, brushwork) in one .relief file. Open it later in Setup and carry on without running the AI again. The browser's autosave only keeps the latest session, so save a project for anything you want to keep.

WHEN THINGS GO WRONG

Quick fixes

PROBLEM	TRY THIS
Background is part of the relief	Background & mask: set Mask to Auto or Flat colour, or paint Remove over it.
Not sure which preset	Preset: **Decide for me…**
Whole thing looks tilted	Shape: raise Auto-level tilt and Level field.
Grainy or noisy surface	Finish: raise Denoise, lower Fine detail. Use a sharper picture.
Waffle pattern on flat areas	Generate: make sure **Remove patch grid** is on. Lower Normal detail if blotches remain.
Too flat, no depth	Use **Anchored · keeps volume**, or lower Compress depth.
Coin rim uneven	Edges: **Rim lock**, double-click the slider.
Hard cliff around the edge	Edges: raise Edge roll-off.
“Device lost” / GPU error	Press **Recover GPU**, then Run passes again. Close heavy programs, lower Normal detail.
Browser crashed	Reopen the page: Setup → **Restore last session**.
Very slow	Use a smaller Working size and fewer tiles, and keep Fast preview on.
Changed something by mistake	↶ **Undo** (Ctrl+Z). Messed it all up: **Reset all settings**.

THE SMALL PRINT

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Relief Forge uses these third-party parts, each under its own licence:

COMPONENT	BY	LICENCE
Depth Anything 3 · Mono Large	ByteDance Seed (Depth Anything team)	Apache 2.0
DA3 Mono Large ONNX conversion	Jens Duttke	Apache 2.0
Depth Anything V2 · Large / Base optional	Depth Anything team; ONNX by onnx-community	CC BY-NC 4.0 · **non-commercial use only**
MoGe-2 normal models	Microsoft	MIT
DINOv2 (inside MoGe-2)	Meta AI	Apache 2.0
ONNX Runtime Web	Microsoft	MIT
three.js (3D view)	three.js authors	MIT
Manrope, IBM Plex Mono (fonts)	Mikhail Sharanda; IBM	SIL Open Font Licence 1.1

Commercial work: if you sell what you make, use Depth Anything 3 for depth. Depth Anything V2 Large and Base are for non-commercial use only.

The AI models are downloaded from Hugging Face the first time you use them and are not part of the Relief Forge app itself. Full licence texts are on each project's page. This summary is not legal advice.